

BRENNAN J. SULLIVAN, PHD
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Education

Baylor College of Medicine Ph.D. – Neuroscience Advisor: Jeff Magee	2021 - 2026
Rhodes College B.S. – Neuroscience Advisor: Kelly Dougherty	2013 - 2017

Research Positions

Graduate Student Jeff Magee Lab Howard Hughes Medical Institute Duncan Neurological Research Institute, Texas Children's Hospital Department of Neuroscience, Baylor College of Medicine	2022 - 2026
Research Assistant Shilpa Kadam Lab Hugo W. Moser Research Institute, Kennedy Krieger Children's Hospital Department of Neurology, Johns Hopkins University School of Medicine	2017 - 2021
Undergraduate Research Fellow Kelly Dougherty Lab Program in Neuroscience Department of Biology, Rhodes College	2016 - 2017

Publications

Articles [* denotes equal contribution]

1. Xiao K.*, Li Y.*, **Sullivan B.J.***, Li. G*, Magee J.C. (2025) Rapid neocortical network modifications via dendritic plateau potential induced plasticity. *bioRxiv*. DOI:10.1101/2025.11.19.689338.
2. Thomas A.*, Yang W.*, Wang C., Tipparaju S.L., Chen G., **Sullivan B.J.**, Swiekatowski K., Tatam M., Gerfen C., Li N. (2023) Superior colliculus cell types bidirectionally modulate choice activity in frontal cortex. *Nature Communications* PMID: 37963894
3. **Sullivan B.J.**, Kipnis P.K., Carter B.M., and Kadam S.D. (2021) Targeting ischemia-induced KCC2 hypofunction rescues refractory neonatal seizures and mitigates epileptogenesis in a mouse model. *Science Signaling*. PMID: 34752143
4. **Sullivan B.J.**, Ammanuel S., Kipnis P.K., Araki Y., Hukanir R.L., and Kadam S.D. (2020) Low-dose Perampanel rescues cortical gamma dysregulation associated with parvalbumin interneuron GluA2 upregulation in epileptic Syngap1^{+/-} mice. *Biological Psychiatry*. PMID: 32107006
5. Kipnis P.K.*, **Sullivan B.J.***, Carter B.M., and Kadam S.D. (2020) TrkB agonists prevent post-ischemic BDNF-TrkB mediated emergence of refractory neonatal seizures in CD-1 pups. *JCI Insight*. PMID: 32427585.

6. Kang J., Kadam S.D., Elmore J.S., **Sullivan B.J.***, Valentine H.*, Malla A.P., Grace A.A., Rahmim A., Loew L.M., Baumann M., Gjedde A., Boctor E.M., and Wong D.F. (2020) Transcranial photoacoustic imaging of NMDA-evoked focal circuit dynamics in rat forebrain. *Journal of Neural Engineering*. PMID: 32084654.
7. Carter B.M., **Sullivan B.J.**, Landers J.R., Kadam S.D. (2018) Dose-dependent reversal of KCC2 hypofunction and phenobarbital-resistant neonatal seizures by ANA12. *Scientific Reports*. PMID: 30097625

Reviews

1. **Sullivan B.J.** and Kadam S.D. (2021) Brain Derived Neurotrophic Factor in Neonatal Seizures. *Pediatric Neurology*. PMID: 33773288
2. Kadam S.D., **Sullivan B.J.**, Goayal A., Blue M.E., and Smith-Hicks C. (2020) Rett Syndrome and CDKL5 Deficiency Disorder: From Bench to Clinic. *International Journal of Molecular Sciences*. PMID: 31618813
3. Kipnis P.K., **Sullivan B.J.**, and Kadam S.D. (2019) Sex Dependent Signaling Pathways Underlying Seizure Susceptibility and the Role of Chloride Cotransporters. *Cells*. PMID: 31085988

Chapters

1. **Sullivan B.J.** and Kadam S.D. (2021) Protocol for Drug Screening with Quantitative Video Electroencephalography in a Translational Model of Refractory Neonatal Seizures. In *Experimental and Translational Models to Screen Drugs Effective Against Seizures and Epilepsy*. D. Vohora, Ed. Springer, New York
2. **Sullivan B.J.** and Kadam S.D. (2020) The involvement of neuronal chloride transporter deficiencies in epilepsy. In *Neuronal Chloride Transporters in Health and Disease*. X. Tang, Ed. Elsevier, Amsterdam, Netherlands.

Invited Talks

Rush and Helen Record Neuroscience Forum, Galveston, TX Baylor College of Medicine Neuroscience Department Annual Retreat <i>Mechanisms of neuronal feature selectivity in the lateral visual cortex.</i>	2025
Johns Hopkins University and Kennedy Krieger Institute, Baltimore, MD Pediatric Neuroscience Collaboration Seminar, Virtual Live Content Event <i>Hitting a moving target: KCC2 and pharmacoresistant seizures in the developing brain.</i>	2022
3rd International SYNGAP1 Conference Young Investigator Workshop, Virtual Live Content Event <i>An early forecast for the SYNGAP1 storm.</i>	2021
14th European Congress on Epileptology, Geneva, Switzerland Platform session, Cancelled due to COVID-19 <i>The KCC2 functional enhancer CLP290 rescues phenobarbital-resistant neonatal seizures.</i>	2020
The Scripps Research Institute, Jupiter, Florida 2nd International SYNGAP1 Conference <i>Perampanel rescues behavioral state-dependent transitions on qEEG in Syngap1^{+/-} mice.</i>	2018

Rhodes College, Memphis, TN Summer Undergraduate Research Symposium <i>A hitchhikers guide to voltage-gated K⁺ channels.</i>	2016
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Awards

NINDS Curing the Epilepsies Conference Travel Award <i>CURE Epilepsy</i>	2021
14th European Congress on Epileptology Travel Award <i>International League Against Epilepsy</i>	2020
American Epilepsy Society Annual Meeting Travel Award <i>International League Against Epilepsy</i>	2019
Young Investigator Travel Award <i>Syngap1 Foundation</i>	2019
Undergraduate Research Fellowship <i>Rhodes College</i>	2016

Service

Ad-Hoc reviewer for <i>eLife</i> , <i>BRAIN</i> , <i>Molecular Psychiatry</i> , <i>Journal of Neural Engineering</i> , <i>Epilepsy Research</i> , and <i>Neuropharmacology</i>	
Baylor College of Medicine Neuroscience Program Bootcamp <i>Lecturer and Volunteer</i>	2022 - 2025
NINDS-AES Young Investigators Virtual Seminar <i>Seminar Co-Organizer, Emerging neurotechnology to disrupt epilepsy</i>	2021
Johns Hopkins Internship in Brain Sciences <i>Mentor</i>	2018 - 2020